

# Mangalam HDPE Pipes

Premium Infrastructure Solutions — Product Landing Page

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HTML5

Vanilla CSS

Vanilla JS

Responsive

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## 1. Project Overview

This project is a high-fidelity, responsive product landing page for **Mangalam HDPE Pipes** — a B2B manufacturer of premium HDPE pipes, coils, and fittings for infrastructure applications. The page is designed as a Gushwork frontend assignment to demonstrate modern UI/UX skills using only vanilla web technologies (no frameworks).

**Goal:** Build a pixel-perfect, production-grade product page with rich interactions, smooth animations, and full mobile responsiveness — all without external UI libraries or CSS frameworks.

## 2. Technology Stack

| Layer      | Technology                | Notes                                           |
|------------|---------------------------|-------------------------------------------------|
| Structure  | HTML5                     | Semantic elements, accessible markup            |
| Styling    | Vanilla CSS3              | CSS variables, Flexbox, Grid, custom animations |
| Logic      | Vanilla JavaScript (ES6+) | No jQuery, no external libraries                |
| Typography | Google Fonts (Inter)      | Weights: 400, 500, 600, 700                     |
| Icons      | Inline SVG                | No icon library dependency                      |

## 3. File Structure

```
GushWork_Assignment_NagarjunG/
├─ index.html           (713 lines – Main page markup)
├─ styles.css           (965 lines – Complete styling)
├─ script.js            (190 lines – All interactive logic)
├─ documentation.html   (This file)
├─ assets/
│   ├── logo.png
│   └── fishnet.jpg
```

- └─ portfolio1.jpg
- └─ portfolio2.jpg
- └─ euroflex.png
- └─ Vector.png
- └─ BIS-Hallmark-Color.png
- └─ CE Certification.png
- └─ Frame 2147224846.png

## 4. Page Sections & Features

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### 4.1 Header & Navigation

- Fixed brand logo with desktop navigation links
- Yellow accent line for visual branding
- **Sticky header clone:** A duplicate header appears with a glass-blur effect after scrolling 80% of the viewport height

### 4.2 Hero Section

- **Image Gallery:** Main product image with 6 thumbnails and prev/next navigation
- **Hover Zoom:** Mouse-tracking 2× zoom on the main image using transform-origin
- Certification badges (BIS, ISO, CE) with icon images
- Price card with shipping/return info badges
- Two CTA buttons: "Get Custom Quote" (opens modal) and "View Technical Specs"

### 4.3 Trust Bar

A row of grayscale partner logos demonstrating global client trust.

### 4.4 Technical Specifications

- Dark-themed section with a styled data table (10 rows)
- Parameters: Diameter, Pressure Ratings, SDR, Temperature, Service Life, Density, Certifications, Joint Type, Coil Lengths, Origin
- "Download Datasheet" button triggers the download modal

### 4.5 Benefits Section

A 3-column grid of 6 benefit cards, each with an SVG icon, title, and description. Cards feature hover elevation animations.

### 4.6 FAQ Accordion

- 5 collapsible FAQ items with smooth toggle
- Only one item can be open at a time (auto-collapse others)
- Catalogue email promo box with inline form below the FAQ

## 4.7 Industries Carousel

4-column image card grid with gradient overlays, previous/next navigation buttons, and an "active" card highlight with a green border.

## 4.8 Manufacturing Process Tabs

- 8 horizontal tab buttons in a pill-style navigation bar
- Tab content updates dynamically (title, description, image) with a fade transition
- Horizontally scrollable on mobile devices

## 4.9 Testimonials

4-column grid of testimonial cards with quote icons, review text, and author metadata (avatar, name, role). Hover lift effect.

## 4.10 Portfolio Grid

3-column product portfolio with cards containing title, description, image, and "Learn More" link. Followed by an "expert CTA" box with phone icon.

## 4.11 Resources & Downloads

Styled list of 3 downloadable PDF resources with hover highlight.

## 4.12 Contact CTA Section

Full-width blue CTA box with a two-column layout: promotional copy on the left, and a white form card on the right with name, company, email, and phone fields.

## 4.13 Footer

- Branded promo bar with logo
- 4-column link grid: About, Categories, Products, Contact
- Social media links (LinkedIn, Twitter/X, Instagram)
- Bottom bar with copyright and legal links

## 4.14 Modals

- **Download Modal:** Email + optional phone to request catalogue
- **Quote Modal:** Full name, company, email, phone for callback request
- Both use backdrop blur, slide-up animation, and close on outside click

## 5. CSS Architecture

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### 5.1 Design Tokens (CSS Custom Properties)

```
:root {  
  --primary-blue: #25419A;  
  --accent-blue: #3D5AFE;  
  --dark-bg: #0B0E14;  
  --light-bg: #F8F9FA;  
  --border-color: #EAEAEA;  
  --text-main: #1A1A1A;  
  --text-muted: #666666;  
  --white: #FFFFFF;  
  --yellow-highlight: #FEF9C3;  
  --yellow-text: #854D0E;  
  --container-max: 1200px;  
}
```

### 5.2 Layout Strategy

- **Container:** Max-width 1200px, centered with auto margins, 40px horizontal padding
- **Grid:** Used for hero (2-col), benefits (3-col), industries (4-col), testimonials (4-col), portfolio (3-col), footer (4-col)
- **Flexbox:** Used for navigation, badge rows, action buttons, form layouts

### 5.3 Animation System

- **Scroll reveal:** .fade-up class with IntersectionObserver (opacity 0 → 1, translateY 30px → 0)
- **Hover effects:** translateY(-5px) elevation on cards with box-shadow transitions
- **Sticky header:** CSS top transition from -100px → 0 with backdrop-filter blur
- **Modal:** Opacity + translateY transition on overlay and content

### 5.4 Total CSS Size

965 lines / ~28 KB — organized sequentially by section (header → hero → trust → specs → benefits → FAQ → industries → process → testimonials → portfolio → resources → CTA → footer → modals → media queries).

## 6. JavaScript Functionality

All logic is contained in a single `script.js` (190 lines), wrapped in a `DOMContentLoaded` listener.

| Feature           | Implementation                                                                                                              |
|-------------------|-----------------------------------------------------------------------------------------------------------------------------|
| Hero Gallery      | <code>updateHeroImage(index)</code> — swaps main image src and active thumbnail class. Prev/next buttons cycle with modulo. |
| Process Tabs      | Data-driven tab switching with a <code>tabData</code> object. Content swaps with a 300ms opacity fade.                      |
| FAQ Accordion     | Click handler on headers toggles <code>.active</code> class. Auto-closes sibling items.                                     |
| Sticky Header     | Clones header DOM, appends to body. Scroll listener adds <code>.visible</code> class at 80vh threshold.                     |
| Image Hover Zoom  | Mousemove computes cursor position as percentage, sets <code>transform-origin</code> and <code>scale(2)</code> .            |
| Scroll Animations | IntersectionObserver at 10% threshold adds <code>.visible</code> to <code>[data-aos]</code> elements.                       |
| Download Modal    | Open/close with class toggle + body overflow lock. Closes on backdrop click.                                                |
| Quote Modal       | Same pattern as download modal with <code>preventDefault()</code> on button click.                                          |

## 7. Responsive Design Strategy

Three breakpoint tiers with mobile-first progressive enhancement:

| Breakpoint | Target        | Key Changes                                                                    |
|------------|---------------|--------------------------------------------------------------------------------|
| ≤ 1200px   | Small desktop | Container to 95%, hero title reduced                                           |
| ≤ 1024px   | Tablet        | Hero → 1 col, grids → 2 col, CTA → 1 col, footer → 2 col                       |
| ≤ 768px    | Mobile        | All grids → 1 col, stacked header, smaller typography, horizontal table scroll |

**Tab Navigation:** The process tabs use overflow-x: auto with hidden scrollbars for horizontal swipe on mobile.

## 8. Design System & Tokens

### Color Palette

- **Primary:** #25419A (Deep Blue)
- **Accent:** #3D5AFE (Vivid Blue)
- **Dark BG:** #0B0E14
- **Light BG:** #F8F9FA / #F9FAFB
- **Text:** #1A1A1A / #666666
- **Borders:** #E5E7EB / #EAEAEA
- **Yellow Badge:** #FEF9C3 bg / #854D0E text
- **Success:** #10B981 (Green border)

### Typography

- **Font:** Inter (Google Fonts)
- **H1:** 3.2rem / 700 weight
- **H2:** 2.2rem / 700 weight
- **H3:** 1.1–1.8rem / 600–700
- **Body:** 0.9rem / 400 weight
- **Small:** 0.8–0.85rem / 500–600
- **Line-height:** 1.5–1.7

### Spacing

### Effects



- Container padding: 40px
- Section padding: 60px vertical
- Card padding: 32px–48px
- Grid gaps: 20px–60px
- Border radius: 6px–32px

- Card shadows: 0 4px 20px rgba(0,0,0,.05)
- Hover shadows: 0 10px 30px rgba(0,0,0,.05)
- Backdrop blur: blur(10px) on sticky header
- Modal blur: blur(4px) on overlay
- Transitions: 0.2s–0.3s ease

## 9. How to Run

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No build process or dependency installation is required. This is a static site.

```
# Option 1: Open directly in browser
open index.html

# Option 2: Use a local server (recommended)
npx serve .
# Then visit http://localhost:3000

# Option 3: VS Code Live Server
# Right-click index.html → "Open with Live Server"
```

**No dependencies.** The project has zero npm packages. The only external resource is the Google Fonts CDN link for the Inter typeface.